Q1: Loading and Understanding the Dataset (10) marks

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
#To suppress all warnings
import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
df = sns.load_dataset('tips')
df.head(5)
```

Out[2]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

In [3]:
```
df.columns
```

Out[3]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dty
pe='object')
```

In [4]:
```
df.shape
```

Out[4]:
```
(244, 7)
```

Q2: Basic Filtering – Who Tips More? (25 marks)

In [5]:
```
weekend = df[df['day'].isin(['Sat', 'Sun'])]
```

In [6]:
```
weekdays = df[df['day'].isin(['Thur', 'Fri'])]
```

In [7]:
```
tip1 = weekend['tip'].mean()
print(tip1)
```

```
3.115276073619632
```

In [8]:
```
tip2 = weekdays['tip'].mean()
print(tip2)
```

```
2.7628395061728392
```

In [9]:
```
total1 = weekend['total_bill'].mean()
```

In [26]:
```
total2 = weekdays['total_bill'].mean()
```

Answer Q2.3: Compare this with weekday customers

Weekend(3.1 avg) customers are more tip-friendly, whereas weekdays(2.7 avg) customers are not,

Q3: Sorting and Identifying High-Value Customers(15 marks)

In [30]:
```
sort = df.sort_values(by='total_bill', ascending=False)
```

In [31]:
```
sort.head(10)
```

Out[31]:

	total_bill	tip	sex	smoker	day	time	size
170	50.81	10.00	Male	Yes	Sat	Dinner	3
212	48.33	9.00	Male	No	Sat	Dinner	4
59	48.27	6.73	Male	No	Sat	Dinner	4
156	48.17	5.00	Male	No	Sun	Dinner	6
182	45.35	3.50	Male	Yes	Sun	Dinner	3
102	44.30	2.50	Female	Yes	Sat	Dinner	3
197	43.11	5.00	Female	Yes	Thur	Lunch	4
142	41.19	5.00	Male	No	Thur	Lunch	5
184	40.55	3.00	Male	Yes	Sun	Dinner	2
95	40.17	4.73	Male	Yes	Fri	Dinner	4

3. What do you notice about their group sizes?

The majority of groups are small 2-3 people, the bigger group size means, more tips

Q4: Data Quality Check & Cleaning (10 marks)

In [32]:
```
print(df.isnull().sum())
```

```
total_bill    0
tip           0
sex           0
smoker        0
day           0
time          0
size          0
dtype: int64
```

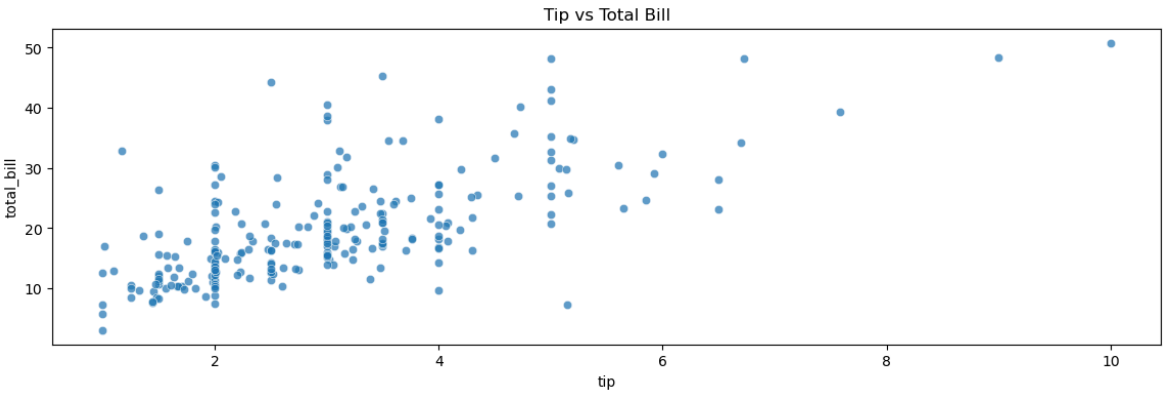
Answer Q4.2: Why missing values are dangerous in machine learning.

Missing values are making analysis more unaccurate, computer is trying to learn from examples, and when some information is missing, the examples become incomplete

Q5: Visualization – What Influences Tips? (25 marks)

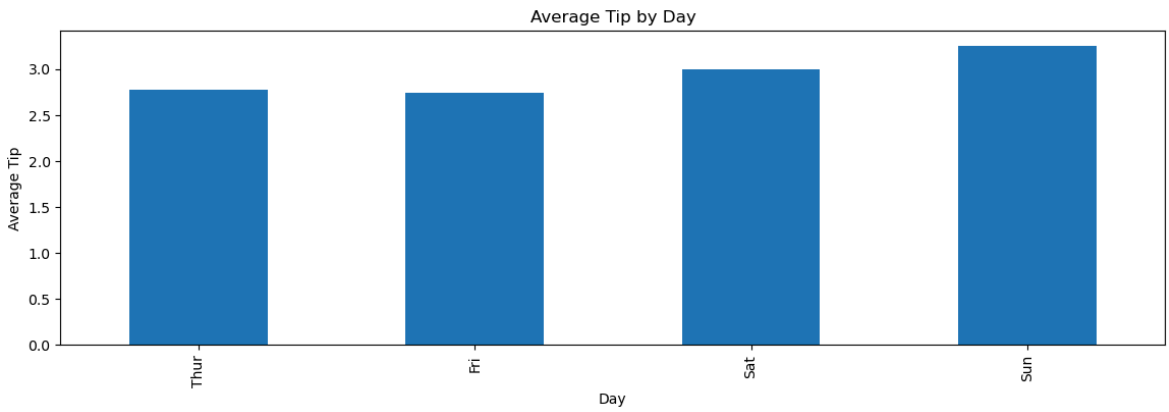
In [14]:
```
plt.figure(figsize=(14, 4))

sns.scatterplot(x='tip', y='total_bill',
alpha=0.7, data=df)
plt.title('Tip vs Total Bill')
plt.show()
```



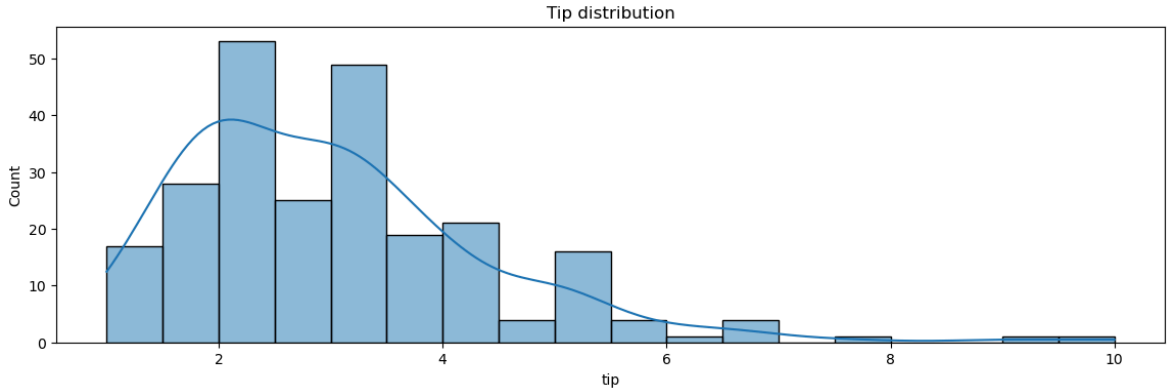
In [15]:
```
plt.figure(figsize=(14, 4))

df.groupby('day')['tip'].mean().plot(kind='bar')
plt.title('Average Tip by Day')
plt.xlabel('Day')
plt.ylabel('Average Tip')
plt.show()
```



In [42]:
```
plt.figure(figsize=(14, 4))
sns.histplot(df['tip'], kde=True)
plt.title('Tip distribution')
```

Out[42]: Text(0.5, 1.0, 'Tip distribution')



Answer Q5.2: Interpret each plot in 1–2 sentences.

In the first plot, we can see that the highest amount of tips is received on Sunday, then Saturday, which means that tips are mostly given on weekends.

In the second plot, we can see that people most often give tips between $2 and 4$, while tips between $4 and 10$ are less common.

Q6: Final Insight & Conclusion (25 marks)

Customers who come on weekends give more tips than on other days.

Customers on Sunday give the highest tips, with Saturday slightly less.

Customers are more tip-friendly on weekends, which means managers could offer more attractive orders, manage resources more efficiently, and create special programs that could increase income.

Answer Q6:

Customers – Those who visit on weekends give higher tips than on other days, with the highest on Sunday and slightly lower on Saturday.

Restaurant Managers – Since customers are more generous on weekends, managers can offer more appealing menu options, optimize staffing and resources, and implement special programs to boost revenue.

In [ ]:

```

```